

# PKU–Zurich PhD Summer School on Machine Learning for Macroeconomics and Finance

Overlapping generations, borrowing constraints, product-form KKT  
complementarity

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Based on Azinovic, Gaegauf & Scheidegger (2022, *IER* 63(4), 1471–1525) and Krueger & Kübler (2004, *JEDC*).

[https://liboecon.com/2026\\_summer\\_school.html](https://liboecon.com/2026_summer_school.html)

# Roadmap

## Part I — What is an OLG model?

- ▶ Why overlapping generations; cohort structure; the DEQN recipe in one line.

## Part II — The analytic 6-agent model (Krueger & Kübler, 2004)

- ▶ Model  $\rightarrow$  equilibrium  $\rightarrow$  cost  $\rightarrow$  FREE  $\rightarrow$  architecture  $\rightarrow$  training  $\rightarrow$  validation against the closed form.
- ▶ Primary notebook: `04_02_OLG_Analytic_DEQN_persistent.ipynb`; companion ablation `04_01_OLG_Analytic_DEQN_exogenous.ipynb`.

## Part III — The leading example of AGS (2022): the 56-cohort benchmark

- ▶ Same 9-step recipe at research scale: two assets, two borrowing constraints, capital adjustment costs, persistent shocks, hump-shaped labor.
- ▶ Primary notebook: `04_04_OLG_Benchmark_DEQN_persistent.ipynb`; companion ablation `04_03_OLG_Benchmark_DEQN_exogenous.ipynb`.

# Learning objectives

**By the end of this lecture you should be able to:**

- ▶ Construct a finite-dimensional overlapping-generations equilibrium with cohort-specific savings and lifecycle heterogeneity
- ▶ Derive equilibrium conditions and build the mean-squared Euler-residual loss for a multi-cohort OLG model
- ▶ Implement the DEQN recipe: model specification, equilibrium definition, cost function, network training, and validation
- ▶ Validate a learned solution against the closed-form equilibrium in the analytic Krueger–Kuebler model
- ▶ Scale the method from six cohorts to research benchmarks with capital adjustment costs, multiple assets, and binding constraints

# Why Overlapping Generations?

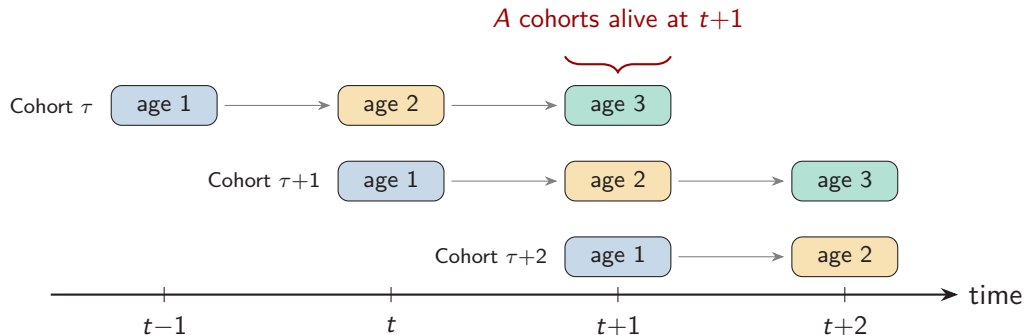
## The economic content

- ▶ **Lifecycle:** agents are born, work, save, retire, die. A representative infinitely-lived household cannot speak to pensions, retirement, inheritance, or cohort-specific shocks.
- ▶ **Demographics:** at any  $t$ , many cohorts coexist, each with its own asset holdings and labor income.
- ▶ **Finite asset distribution:** with  $A$  cohorts, the cross-sectional wealth distribution is an  $A$ -vector – not an infinite-dimensional object. **Perfect for DEQNs:** state stays finite-dimensional with realistic heterogeneity.

## Two canonical versions we solve today

1.  $A = 6$  analytic model (Krueger & Kübler, 2004): closed form  $\Rightarrow$  ground-truth validation.
2.  $A = 56$  leading-example benchmark (Azinovic, Gaegauf & Scheidegger, 2022, §3): research-scale, no closed form.

# OLG Structure: Overlapping Cohorts



At each period  $t$  we have  $A$  coexisting cohorts. A newborn enters (age 1); the oldest dies (age  $A$ ).

**Key observation:** the number of types is finite ( $A$ ), so the cross-sectional state is finite-dimensional.

# The DEQN Recipe in One Line

From model to neural network (to be applied twice in today's lecture)

1. Write down the **economic model**: preferences, technology, constraints, shocks.
2. Define the **equilibrium**: choices + prices that satisfy optimization + market clearing.
3. Write the **equilibrium conditions** as a system  $G(x, f(x)) = 0$ .
4. Replace  $f$  by a **neural network**  $\mathcal{N}_\theta$  and form the **cost function**  
 $\ell(\theta) = \frac{1}{|D|} \sum_j \|G(x_j, \mathcal{N}_\theta(x_j))\|^2$ .
5. Sample training states  $D$ , minimize  $\ell(\theta)$  by SGD.

## The only thing that depends on the model

The **equilibrium conditions**  $G$ . Everything else – the network, the SGD loop, the sampling method – is generic. **So to solve a new model, write down its  $G$ .**

Today's deliverable: derive  $G$  explicitly for both the analytic and the benchmark OLG models.

# II.1 Setup — the Analytic Model

- ▶ Time discrete:  $t = 0, 1, 2, \dots, \infty$ .
- ▶ Agents live for  $A = 6$  periods.
- ▶ **One representative household per cohort**; a new one is born every period.
- ▶ **No uncertainty about lifetime**: all agents live exactly  $A$  periods.
- ▶ Exogenous aggregate shock  $z_t \in \{1, 2, 3, 4\}$  drawn **i.i.d.** with transition  $\pi_{ss'} = 0.25$ .
- ▶ **Log utility**  $u(c) = \ln c$  – chosen because it gives a **closed-form** equilibrium (Krueger & Kübler, 2004).
- ▶ **Labor**: only the youngest cohort works:  $\ell^1 = 1$ ,  $\ell^s = 0$  for  $s \geq 2$ .

## Shocks: $z_t = (\eta_t, \delta_t)$

| $z$ | $\eta(z)$ | $\delta(z)$ | $\pi$ |
|-----|-----------|-------------|-------|
| 1   | 0.95      | 0.5         | 0.25  |
| 2   | 1.05      | 0.5         | 0.25  |
| 3   | 0.95      | 0.9         | 0.25  |
| 4   | 1.05      | 0.9         | 0.25  |

## Calibration

$\alpha = 0.3$ ,  $\beta = 0.7$ ,  $\gamma = 1$  (log);  $\underline{a} = 0$ .

## II.2 Households — Preferences and the Savings Technology

Agents supply their labor **inelastically** for a market wage  $w_t$ .

An agent **currently of age  $s$**  alive at  $t$  maximises remaining time-separable discounted expected lifetime utility ( $\beta < 1$ ):

$$\mathbb{E}_t \left[ \sum_{i=0}^{A-s} \beta^i \ln(c_{t+i}^{s+i}) \right] \quad (1)$$

Households can save a unit of consumption to obtain a unit of capital good next period, denoted  $a_t^s$ .

The savings become **capital in the next period**:

$$a_t^s = k_{t+1}^{s+1}, \quad \forall t, \forall s \in \{1, \dots, A-1\}. \quad (2)$$



## II.3 Households — Constraints, Budget, Boundary

Households **cannot die with debt**. Borrowing is bounded:

$$a_t^s \geq \underline{a}. \quad (3)$$

In the analytic model we set  $\underline{a} = 0$ : **no borrowing** for any cohort.

At time  $t$ , households sell their capital to the firm at market price  $r_t > 0$ . The per-period **budget constraint** of cohort  $s$  reads

$$c_t^s + a_t^s = r_t k_t^s + \ell^s w_t. \quad (4)$$

**Boundary conditions.** Newborns enter with no inheritance, and the oldest cohort dies without saving anything:

$$k_t^1 = 0, \quad a_t^A = 0.$$

## II.4 Firms and Markets

There is a single representative firm with **Cobb–Douglas** production.

TFP  $\eta$  and depreciation  $\delta$  depend on the exogenous shock  $z$  alone (see II.1 table).

Each period, **after the shock has realized**, the firm buys capital and hires labor to maximize profits, taking prices as given.

The stochastic production function is

$$f(K, L, z) = \eta(z) K^\alpha L^{1-\alpha} + K(1 - \delta(z)).$$

There are **competitive spot markets** for the consumption good, capital, and labor.

In the analytic model  $L_t = \sum_{s=1}^A \ell^s = 1$  (only age-1 works), so the firm problem reduces to  $\max_{K \geq 0} \eta(z) K^\alpha - (r_t - (1 - \delta(z)))K$ .

## II.5 Competitive Equilibrium — Definition

### Definition 1 (competitive equilibrium)

Given initial conditions  $z_0$ ,  $\{k_0^s\}_{s=1}^{A-1}$ , a competitive equilibrium is a collection of choices for households  $\{(c_t^s, a_t^s)_{s=1}^A\}_{t=0}^\infty$ , for the representative firm  $(K_t, L_t)_{t=0}^\infty$ , and prices  $(r_t, w_t)_{t=0}^\infty$ , such that:

1. Given  $(r_t, w_t)_{t \geq 0}$ , household choices maximize the lifetime utility (1) subject to the savings-to-capital identity (2), the borrowing limit (3), and the per-period budget (4):

$$a_t^s = k_{t+1}^{s+1}; \quad a_t^s \geq \underline{a}; \quad c_t^s + a_t^s = r_t k_t^s + \ell^s w_t.$$

2. Given  $(r_t, w_t)$ , the firm maximises profits:

$$(K_t, L_t) \in \arg \max_{K, L \geq 0} f(K, L, z_t) - r_t K - w_t L.$$

3. **All markets clear:** for all  $t$ ,  $L_t = \sum_{s=1}^A \ell^s$ ,  $K_t = \sum_{s=1}^A k_t^s$ .

## II.6 Equilibrium Conditions

**Firm FOCs** (closed form from Cobb–Douglas):

$$w_t = (1 - \alpha) \eta(z_t) K_t^\alpha L_t^{-\alpha}, \quad r_t = \alpha \eta(z_t) K_t^{\alpha-1} L_t^{1-\alpha} + (1 - \delta(z_t)).$$

**Household optimality** for any cohort of age  $s \in \{1, \dots, A-1\}$  (FOC in  $c_t^s$  gives  $\nu_t^s = u'(c_t^s)$ ; FOC in  $a_t^s$  gives the standard **interior** Euler equation):

$$\underbrace{u'(c_t^s)}_{= 1/c_t^s \text{ since } u=\ln} = \beta \mathbb{E}[u'(c_{t+1}^{s+1}) r_{t+1}]$$

**Terminal cohort.** Age  $A$ : consume everything, save nothing.

**Why no KKT multiplier?** In this calibration ( $\gamma = 1$ ,  $\beta = 0.7$ , only the youngest cohort works) the borrowing constraint  $a_t^s \geq 0$  never binds in equilibrium, so  $\lambda_t^s \equiv 0$  and drops out of the Euler equation, the FREE signature, and the loss. Part III re-introduces multipliers for the binding constraints of the 56-cohort benchmark.

## II.7 The DEQN Cost Function

### Explicit cost function

$$\ell_{D_{\text{train}}}(\theta) := \frac{1}{|D_{\text{train}}|} \frac{1}{A-1} \sum_{x_j \in D_{\text{train}}} \sum_{i=1}^{A-1} (e_{\text{REE}}^i(x_j))^2$$

**Relative Euler-equation residual** (cohort  $i$ ):

$$e_{\text{REE}}^i(x_j) := \frac{\left(u'\right)^{-1}\left(\beta \mathbb{E}\left[r\left(\hat{x}_{j,+}\right) u'\left(\hat{c}^{i+1}\left(\hat{x}_{j,+}\right)\right)\right]\right)}{\hat{c}^i\left(x_j\right)} - 1.$$

**Sampling next state.**  $\hat{x}_{j,+} = (z_+, 0, \hat{a}^{[1:A-1]}(x_j))$  with  $z_+$  drawn from  $\pi$  (newborn enters with  $k^1 = 0$ ).

**Feasibility.**  $a^s \geq 0$  and  $c^s \geq 0$  are enforced by the sigmoid savings-fraction output of II.9 (each  $a^s \in [0, \text{inc}^s]$ ); small soft penalties on negative  $c$  and  $K$  are added in the notebook as numerical backstops but are inactive with this head.

## II.8 Functional Rational Expectations Equilibrium (FREE)

**FREE:** one function that solves the equilibrium conditions at every state

**Signature:**  $f : \{1, 2, 3, 4\} \times \mathbb{R}^A \longrightarrow \mathbb{R}^{A-1}$ , input = state  $(z_t, \mathbf{k}_t)$ , output = savings policies.

**Output decomposition** ( $A - 1$  scalars):

$$f\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}\right) = \underbrace{[a_t^1, a_t^2, \dots, a_t^{A-1}]^\top}_{\text{capital-investment policies}}.$$

Such that, for all cohorts  $s = 1, \dots, A - 1$ , the (interior) Euler equation holds:

$$\beta \mathbb{E} \left[ \frac{r_{t+1} u'(c_{t+1}^{s+1})}{u'(c_t^s)} \right] - 1 = 0 \quad (\text{Euler equation}).$$

Stack these  $A - 1$  conditions as  $G([z_t, \mathbf{k}_t], f([z_t, \mathbf{k}_t])) = \mathbf{0}$ .

$A = 6 \Rightarrow 5$  outputs; extended state of dim  $16 + 4A = 40$  – pure feature engineering. The residual *squared* in the loss is the consumption-equivalent form of II.7 (same zeros).

## II.9 Deep Equilibrium Net — Architecture (Analytic)

$$\mathcal{N}_\theta : \{1, 2, 3, 4\} \times \mathbb{R}^A \rightarrow \mathbb{R}^{A-1}$$

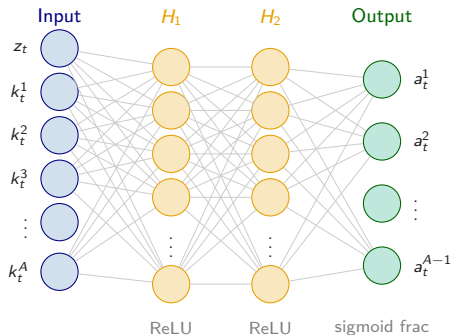
$$\mathcal{N}_\theta\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}\right) = \begin{bmatrix} a_t^1 \\ a_t^2 \\ \vdots \\ a_t^{A-1} \end{bmatrix}$$

such that

$$G\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}, \mathcal{N}_\theta\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}\right)\right) \approx \mathbf{0}.$$

**Sigmoid savings-fraction** output  $\Rightarrow$

$0 \leq a_t^s \leq \text{inc}_t^s$  **by construction**, so  $a_t^s \geq 0$   
(borrowing constraint) and  $c_t^s \geq 0$  both hold  
at every iteration without an explicit multiplier.



2 hidden layers  $\times$  (100, 50) units; single forward pass.

# Companion Notebook Hand-over

Open notebook `04_02_OLG_Analytic_DEQN_persistent.ipynb` (the persistent-simulation primary; the exogenous-sampling companion is `04_01_OLG_Analytic_DEQN_exogenous.ipynb`). It implements the model, equilibrium conditions, loss, FREE map, architecture, training, and validation covered in Part II.

## Cell-to-slide map

| Deck slide                         | Notebook section                                                     |
|------------------------------------|----------------------------------------------------------------------|
| II.1–II.6 Model and equilibrium    | §1–3 parameters, state, network/policy                               |
| II.7–II.9 Loss, FREE, architecture | §4 <code>compute_cost()</code> + training-data construction          |
| II.10–II.13 Training/validation    | §5–8 mini-batches, diagnostics, policy stability, closed-form checks |



## II.10 Sampling and the Training Loop

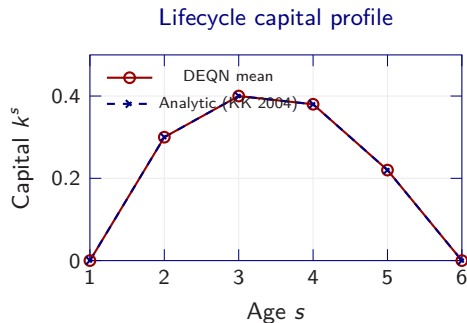
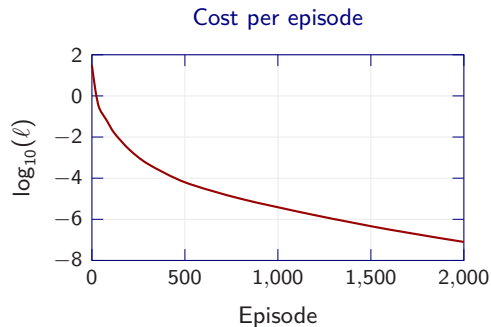
### Cloud method (Azinovic, Gaegauf & Scheidegger, 2022)

1. **Initialisation.** Sample a cloud of initial states  $D_0 = \{x_j^0\}_{j=1}^N$ ; initialise  $\theta^0$ .
2. **Episode  $k$ .**
  - (a) **Simulate.** Draw fresh shocks  $\{z_{+,j}\}$  from  $\pi$ ; roll the cloud forward one period using the **current** policy  $\mathcal{N}_{\theta^k}$ :  $x_j^{k+1} = (z_{+,j}, 0, \mathcal{N}_{\theta^k}^{[1:A-1]}(x_j^k))$ .
  - (b) **Mini-batch SGD.** Take steps Adam steps on  $\ell_{D_k}(\theta)$  using mini-batches from  $D_k$ .
3. **Stop** when  $\log_{10} \ell$  falls below the target.

**Definitions.** **Episode** = one simulate-then-train pass. **Epoch** = one pass through the full cloud.

Sampling from the ergodic set of the current policy makes the cost function focus on states the economy actually visits.

## II.11 Training Results — Analytic Model



### Reading the plots

**Left:** cost falls in  $\log_{10}$  as the network learns the Krueger–Kübler policy; teaching is diagnostic, production reaches tight residuals.

**Right:** trained DEQN savings policy overlays the closed-form Krueger–Kübler solution across cohorts.

# Finger Exercise: Closed-Form Savings Rates

**Setup.** With  $u = \ln$ , cohort  $h$ 's Euler equation reads  $1/c_t^h = \beta \mathbb{E}_t[r_{t+1}/c_{t+1}^{h+1}]$  and the budget constraint is  $c_t^h + a_t^h = \text{inc}_t^h$  where  $\text{inc}_t^h = r_t k_t^h + \ell^h w_t$ . (The household-level setup slides II.2–II.6 used the index symbol  $s$ ; from this slide onward we follow the script convention and use  $h$  instead.)

**Claim (closed-form log-utility OLG savings rate; the validation target used by Krueger & Kübler, 2004).** With log utility and 100% consumption at age  $A$ , the savings rate  $\beta_h := a_t^h / \text{inc}_t^h$  of cohort  $h$  depends only on age:

$$\beta_h = \beta \frac{1 - \beta^{A-h}}{1 - \beta^{A-h+1}}, \quad h = 1, \dots, A-1.$$

**Task (3 minutes).** Compute  $\beta_1, \beta_2, \dots, \beta_5$  with  $\beta = 0.7$ . Interpret:

- ▶ What happens as cohorts approach retirement?
- ▶ Why are younger cohorts saving more as a **rate**?

# Solution: Closed-Form Savings Rates

$$\beta_h = 0.7 \cdot \frac{1 - 0.7^{6-h}}{1 - 0.7^{7-h}}$$

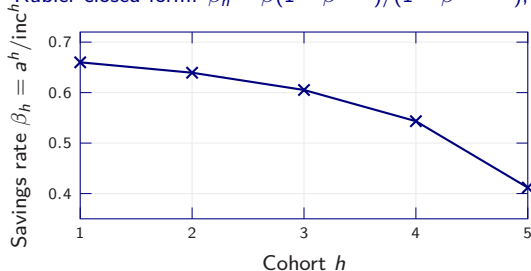
| Cohort $h$ | Periods remaining ( $A - h$ ) | Savings rate $\beta_h$                            |
|------------|-------------------------------|---------------------------------------------------|
| 1          | 5                             | $0.7 \cdot (1 - 0.7^5)/(1 - 0.7^6) \approx 0.660$ |
| 2          | 4                             | $0.7 \cdot (1 - 0.7^4)/(1 - 0.7^5) \approx 0.639$ |
| 3          | 3                             | $0.7 \cdot (1 - 0.7^3)/(1 - 0.7^4) \approx 0.605$ |
| 4          | 2                             | $0.7 \cdot (1 - 0.7^2)/(1 - 0.7^3) \approx 0.543$ |
| 5          | 1                             | $0.7 \cdot (1 - 0.7)/(1 - 0.7^2) \approx 0.412$   |
| 6          | 0                             | 0 (consumes everything)                           |

## Interpretation

- ▶ Savings rate **monotonically declines** with age: as the horizon shrinks, saving less is optimal.
- ▶ Youngest cohort saves  $\sim 66\%$  of its income – longest horizon to enjoy the returns.
- ▶ At  $A$ : terminal boundary  $a^A = 0$  – consume everything.

## II.13 DEQN vs. Analytical Solution

Krueger-Kübler closed form:  $\beta_h = \beta(1 - \beta^{A-h})/(1 - \beta^{A-h+1})$ ,  $\beta = 0.7$ ,  $A = 6$



### Validation strategy

The plotted points are the **exact closed form** from the finger-exercise solution slide. The DEQN should reproduce this curve cohort-by-cohort.

The actual training run, DEQN-vs-analytic comparison, and Euler-error histogram are produced by `04_02_OLG_Analytic_DEQN_persistent.ipynb` (§8). Production target in AGS (2022): mean relative Euler error  $\lesssim 10^{-3}$ .

# II Take-away and Hands-on

## What we just did

1. OLG model with 6 cohorts, log utility, 4 i.i.d. shocks (II.1–II.4).
2. Equilibrium definition + conditions (II.5–II.6).
3. Conditions  $\rightarrow$  **cost function** (II.7).
4. Equilibrium map  $f \rightarrow$  neural network  $\mathcal{N}_\theta$  (II.8–II.9).
5. Trained by sampling + SGD (II.10–II.11), **validated** against the closed form (II.13).

## Hands-on

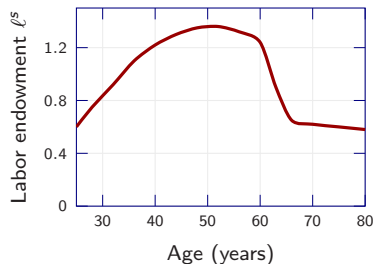
Open `04_02_OLG_Analytic_DEQN_persistent.ipynb`. Set `RUN_MODE = "teaching"` ( $\sim 5$  min) for the classroom run; "smoke" (the default) is a  $\sim 30$ -s sanity check; "production" chases AGS Table 3 accuracy. Both notebooks expose a `SAMPLING_MODE = "simulation" / "exogenous"` switch (the persistent primary uses "simulation", the companion `04_01_OLG_Analytic_DEQN_exogenous.ipynb` uses "exogenous"); the model and loss are identical – only the training cloud differs.

# III.1 Setup — the Leading Example of AGS (2022)

Same recipe as Part II. What changes:

- ▶ Agents live for  $A = 56$  periods (ages 25–80); one period = one year.
- ▶ CRRA utility  $u(c) = c^{1-\gamma}/(1-\gamma)$  with  $\gamma = 2$  (not log).
- ▶ **Hump-shaped labor endowment**  $\ell^s$  (Kotlikoff-style; right panel).
- ▶ **Persistent** Markov shocks on  $(\eta, \delta)$  (not i.i.d.).
- ▶ **Two assets**: capital  $k^s$  and bonds  $b^s$ .
- ▶ **Two constraints**: borrowing  $k'^s \geq 0$  and collateral  $k'^s + \kappa b'^s \geq 0$ .
- ▶ Capital adjustment costs  $\Psi^s = \frac{\zeta}{2}(k'^s - rk^s)^2$ .

Stylised lifecycle labor (schematic)



**Schematic shape only**, rises through working years, drops sharply at retirement, low but nonzero after. See **Brumm, Kübler & Scheidegger (2017)** for the exact  $\ell^s$  used in the notebook; AGS (2022) reuse this calibration as part of their benchmark.

# III.2 Households — Preferences and Savings Technology

An agent of age  $s$  alive at  $t$  maximises

$$\mathbb{E}_t \left[ \sum_{i=0}^{A-s} \beta^i u(c_{t+i}^{s+i}) \right], \quad u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \gamma = 2. \quad (1')$$

**Two saving instruments** (instead of one):

- ▶ **Capital**  $k_{t+1}^{s+1}$ : yields gross return  $r_{t+1}$  next period; incurs an **adjustment cost**  $\Psi_t^s = \frac{\zeta}{2} (k_{t+1}^{s+1} - r_t k_t^s)^2$ .
- ▶ **Bonds**  $b_{t+1}^{s+1}$ : traded at price  $p_t$  today, pay one unit of consumption next period. **Zero net supply**:  $\sum_{s=1}^A b_t^s = 0$ .

Savings become next-period asset holdings ( $\forall s \in \{1, \dots, A-1\}$ ):

$$k_{t+1}^{s+1} = a_{t,k}^s, \quad b_{t+1}^{s+1} = a_{t,b}^s. \quad (2')$$



# III.3 Households — Constraints, Budget, Boundary

**Two constraints on the capital–bonds pair  $(k'^s, b'^s)$ :**

$$k'^s \geq 0 \quad (\text{borrowing constraint}), \quad k'^s + \kappa b'^s \geq 0 \quad (\text{collateral constraint}). \quad (3')$$

**Budget constraint** of cohort  $s$  (bonds enter at price  $p_t$  on the left-hand side, pay off on the right):

$$c_t^s + k_{t+1}^{s+1} + p_t b_{t+1}^{s+1} + \Psi_t^s = r_t k_t^s + b_t^s + \ell_t^s w_t. \quad (4')$$

**Boundary conditions** (born and die without assets):

$$k_t^1 = 0, \quad b_t^1 = 0; \quad k_{t+1}^{A+1} = 0, \quad b_{t+1}^{A+1} = 0.$$

Bond parameter  $\kappa \geq 0$  governs how much bond-short position capital collateralises;  $\kappa = 0$  is a **hard** short-sale constraint on bonds.

# III.4 Firms and Markets — Adding the Bond Market

**Representative firm.** Same Cobb–Douglas as Part II:

$$f(K, L, z) = \eta(z) K^\alpha L^{1-\alpha} + K(1 - \delta(z)).$$

Labor now aggregates over the hump-shaped profile:  $L_t = \sum_{s=1}^A \ell^s$  (a constant).

**Markets cleared in equilibrium.**

1. Labor:  $L_t = \sum_{s=1}^A \ell^s$ .
2. Capital:  $K_t = \sum_{s=1}^A k_t^s$ .
3. Bonds (new):  $0 = \sum_{s=1}^A b_t^s$ . (Zero net supply.)

**Aggregate shocks.**  $z_t \in \{1, \dots, 4\}$  as before, but transition  $\pi^z$  is **persistent** (not i.i.d.):  $(\eta, \delta)$  have their own Markov chains, tensor-producted.

## III.5 Competitive Equilibrium — Definition

### Definition 1' (competitive equilibrium, benchmark)

Given initial conditions  $z_0$ ,  $\{k_0^s, b_0^s\}_{s=1}^{A-1}$ , a competitive equilibrium is a collection of choices  $\{(c_t^s, k_t^{s+1}, b_t^{s+1})_{s=1}^{A-1}\}_{t=0}^{\infty}$ , firm demands  $(K_t, L_t)_{t \geq 0}$ , and prices  $(r_t, w_t, p_t)_{t \geq 0}$ , such that:

1. Households maximize (1') subject to (2')–(4').
2. Firm maximises profits, same closed-form FOCs as Part II.
3. All three markets clear every period: capital, labor, bonds.

Relative to Part II: one new price ( $p_t$ ), one new market-clearing condition (bonds), one new asset class per cohort in the choice vector. Everything else is the same.

## III.6 Equilibrium Conditions (Benchmark)

**Firm FOCs.** Same closed form as Part II.

**Two Euler equations per cohort**, for  $s = 1, \dots, A - 1$  (with the marginal adjustment-cost wedge  $\mathcal{A}_t^s := 1 + \zeta(k_t'^s - r_t k_t^s)$ ):

$$\text{(capital)} \quad u'(c_t^s) \mathcal{A}_t^s = \beta \mathbb{E}[u'(c_{t+1}^{s+1}) r_{t+1} \mathcal{A}_{t+1}^{s+1}] + \lambda_b^s + \lambda_c^s,$$

$$\text{(bonds)} \quad u'(c_t^s) \cdot p_t = \beta \mathbb{E}[u'(c_{t+1}^{s+1})] + \kappa \lambda_c^s.$$

**Two KKT triples per cohort.**

$$\lambda_b^s \geq 0, \quad k'^s \geq 0, \quad \lambda_b^s \cdot k'^s = 0; \quad \lambda_c^s \geq 0, \quad k'^s + \kappa b'^s \geq 0, \quad \lambda_c^s \cdot (k'^s + \kappa b'^s) = 0.$$

**Terminal cohort.** Age  $A$ : consume everything.

So per state we have  $4(A - 1)$  per-agent residuals + 1 bond market-clearing residual =  $4(A - 1) + 1$  equations.

# III.7 The DEQN Cost Function (Benchmark)

Five loss components, one cost function

$$\ell_{D_{\text{train}}}(\theta) = \frac{1}{|D_{\text{train}}|} \frac{1}{4(A-1)+1} \sum_{x_j \in D_{\text{train}}} \left[ \sum_{s=1}^{A-1} R^s(x_j)^2 + (e_{\text{MC},b}(x_j))^2 \right],$$

---

$$\text{where } R^s(x_j)^2 := (e_{\text{REE},k}^s)^2 + (e_{\text{REE},b}^s)^2 + (e_{\text{KKT},b}^s)^2 + (e_{\text{KKT},c}^s)^2.$$

**Component map.**

- ▶  $e_{\text{REE},k}^s, e_{\text{REE},b}^s$ : relative residuals of the capital and bond Euler equations.
- ▶  $e_{\text{KKT},b}^s = \lambda_b^s \cdot k'^s$ : borrowing KKT.
- ▶  $e_{\text{KKT},c}^s = \lambda_c^s \cdot (k'^s + \kappa b'^s)$ : collateral KKT.
- ▶  $e_{\text{MC},b} = \sum_{s=1}^{A-1} \hat{b}'^s$ : bond market clearing (**chosen** next-period bonds, not the current state).

With  $A = 56$  this is  $4 \times 55 + 1 = 221$  loss terms per training state. The structure is the same as Part II: squared equilibrium residuals.

# III.8 Functional Rational Expectations Equilibrium

FREE for the benchmark

$$f : \{1, \dots, 4\} \times \mathbb{R}^A \times \mathbb{R}^A \longrightarrow \mathbb{R}^{4(A-1)+1}$$

$$f\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \\ \mathbf{b}_t \end{bmatrix}\right) = \begin{bmatrix} \hat{k}_t'^1, \dots, \hat{k}_t'^{A-1} \\ \hat{\lambda}_{b,t}^1, \dots, \hat{\lambda}_{b,t}^{A-1} \\ \hat{b}_t'^1, \dots, \hat{b}_t'^{A-1} \\ \hat{\mu}_t^1, \dots, \hat{\mu}_t^{A-1} \\ \hat{p}_t \end{bmatrix}^\top$$

Output blocks: capital savings; borrowing multipliers; bond holdings  $\hat{b}^h$  (transformed to satisfy the collateral floor  $\hat{k}^h + \kappa \hat{b}^h \geq 0$ ); collateral multipliers; equilibrium bond price (one scalar).

**Such that**  $G([z_t, \mathbf{k}_t, \mathbf{b}_t], f([z_t, \mathbf{k}_t, \mathbf{b}_t])) = \mathbf{0}$  where  $G$  stacks the  $4(A-1) + 1$  equations of III.6.

With  $A = 56$ : minimal state dim = 113; the notebook input is 240 after engineered aggregates and transition probabilities. Output dim =  $4 \times 55 + 1 = 221$ . Collateral is then formed as  $\hat{q}^h = \hat{k}^h + \kappa \hat{b}^h$ ; the output transforms enforce  $\hat{k}^h, \hat{\lambda}_b^h, \hat{\mu}^h \geq 0$  and the collateral floor  $\hat{q}^h \geq 0$  (via  $\hat{b}^h \geq -\hat{k}^h/\kappa$ ).

## III.9 Architecture — Benchmark

$$\mathcal{N}_\theta : \mathbb{R}^{240} \rightarrow \mathbb{R}^{221}$$

Sizes from 04\_04\_OLG\_Benchmark\_DEQN\_persistent.ipynb

|                |                                                                   |
|----------------|-------------------------------------------------------------------|
| Notebook input | 240 (minimal state: 113)                                          |
| Hidden layers  | $2 \times 128$ (teach.) / $2 \times 1000$ (prod.)                 |
| Output dim     | $4(A-1) + 1 = 221$                                                |
| Output layer   | softplus (multipliers) / bounded-exp (capital, bonds, bond price) |
| Optimizer      | Adam, lr $10^{-5}$                                                |

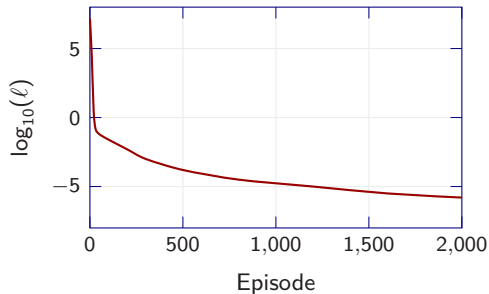
### Output slicing

$$\mathcal{N}_\theta(\mathbf{x}_t) = (\hat{k}'^{1:A-1}, \hat{\lambda}_b^{1:A-1}, \hat{b}'^{1:A-1}, \hat{\mu}^{1:A-1}, \hat{p}).$$

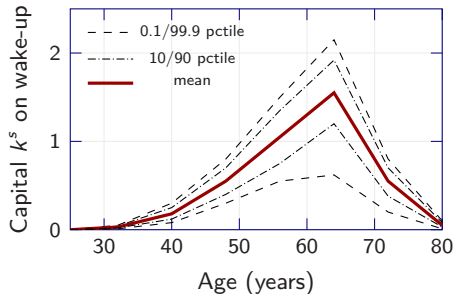
Same MLP template as Part II, plus a bond-price head; collateral formed as  $\hat{q}^h = \hat{k}'^h + \kappa \hat{b}'^h$ .

# III.10 Training — Loss and Lifecycle Capital

Cost per episode (teaching,  $128^2$ , 2,000 ep.)



Lifecycle capital



## Reading the plots

**Left:** loss falls sharply but oscillates because each episode re-simulates the cloud.

**Right:** lifecycle capital is hump-shaped: accumulation during working years, decumulation after retirement. Bands are simulated percentiles.



# III.11 Consumption Profiles by Aggregate State

## State-conditioned slices of the trained policy

For each aggregate state  $z \in \{1, 2, 3, 4\}$  the trained policy yields a lifecycle consumption profile  $c^s(z)$  over ages  $s = 25, \dots, 80$ , read off a long ergodic simulation from `04_04_OLG_Benchmark_DEQN_persistent.ipynb`. (The notebook ships pooled lifecycle bands and residual diagnostics; the per-shock breakdown is a small add.)

## The four aggregate states

| $z$ | $\eta(z)$ | $\delta(z)$ | expected consumption ranking                       |
|-----|-----------|-------------|----------------------------------------------------|
| 2   | 1.022     | 0.08        | <b>highest</b> – best technology, low depreciation |
| 4   | 1.022     | 0.11        | high                                               |
| 1   | 0.978     | 0.08        | low                                                |
| 3   | 0.978     | 0.11        | <b>lowest</b> – bad technology, high depreciation  |

## What to look for in the live plot

Four distinct profiles, ordered as in the table; exact magnitudes and crossover ages come from §11 of the notebook.

# III.12 Reading the Diagnostics

## How do we know the network actually solved the equilibrium?

1. **Loss level.**  $\log_{10} \ell \lesssim -5$  means equilibrium residuals at the  $10^{-3}$  level or smaller – AGS (2022) threshold.
2. **Per-equation Euler errors.** Plot mean  $|e_{\text{REE},k}^s|$  and  $|e_{\text{REE},b}^s|$  by age  $s$ ; expect  $\sim 10^{-3}$  everywhere.
3. **Bond market clearing.**  $|e_{\text{MC},b}| < 10^{-4}$  on the ergodic simulation path.
4. **Borrow-binding fraction.** Ages where  $\lambda_b^s > 0$ ; standard calibration: only the youngest few cohorts.
5. **Policy-drift on a fixed anchor cloud.** Evaluate the policy on `X_anchor` each monitoring interval; report `policy_drift_rms`, `policy_drift_max`; flag the run *time-invariant* once both fall below `TIME_INVARIANCE_TOL_RMS / _MAX`.
6. **Ergodic simulation.** Simulate 10 000 periods; plot lifecycle consumption + capital with percentile bands.

**Hands-on:** `04_04_OLG_Benchmark_DEQN_persistent.ipynb` (persistent-simulation primary; `04_03_OLG_Benchmark_DEQN_exogenous.ipynb` is the feedback-free ablation, same model and loss, different training cloud). §1–4 match III.1–III.5; §5–7 = training and policy stability; §8 = plots/diagnostics of III.10–III.12. Teaching runtime ( $\sim 200$  segments, 128-wide): a few minutes.

# Summary

## What we accomplished

1. Introduced OLG as a finite-dimensional family of heterogeneous-agent models – well suited to DEQNs because the aggregate state stays finite-dimensional.
2. Applied the DEQN recipe **twice**:
  - ▶ **Part II**: the 6-agent Krueger–Kübler (2004) analytic model, validated against the closed form ( $\lesssim 10^{-3}$  mean Euler error).
  - ▶ **Part III**: the 56-cohort leading-example benchmark of AGS (2022): CRRA, hump labor, persistent shocks, two assets, two borrowing constraints, capital adjustment costs.
3. **Same 9-step recipe** both times: setup, households I/II, firms, equilibrium, conditions, cost, FREE, architecture.

## One line to remember

Writing down a DEQN = writing down the equilibrium conditions  $G$ . Everything else (network, SGD, sampling) is boilerplate.

**Hands-on exercise:** 04\_05\_OLG\_Exercise.ipynb — guided tasks on the analytic 6-agent model, building on notebook 04\_02.

# References and Further Reading

- ▶ Azinovic, M., Gaegauf, L., & Scheidegger, S. (2022). *Deep equilibrium nets*. International Economic Review 63(4), 1471–1525. [notebook 04\_04 (benchmark\_persistent) implements §3 of this paper].
- ▶ Krueger, D., & Kübler, F. (2004). *Computing equilibrium in OLG models with stochastic production*. Journal of Economic Dynamics & Control 28(7), 1411–1436. [notebook 04\_02 (analytic\_persistent) implements their 6-agent analytic model].
- ▶ Fischer, A. (1992). *A special Newton-type optimization method*. Optimization 24, 269–284. [FB-complementarity function – used in **Session 3 (IRBC)**; mentioned in this lecture only as an alternative to the product-form KKT residuals adopted here.]
- ▶ Code: <https://github.com/sischei/DeepEquilibriumNets>.

# Optional Next: Agentic-Programming Workshop

**What's next (optional, no lecture-script chapter).** A hands-on workshop in which an LLM-driven coding agent (Claude Code, Cursor, Codex) acts as a pair programmer on a fresh DEQN model.

**Coming up tomorrow (Day 3):** surrogate models & Gaussian processes (incl. Bayesian active learning), structural estimation with simulated method of moments, and deep uncertainty quantification & optimal policy. *Beyond this course:* heterogeneous agents on a histogram (Young's method), continuum-of-agents DEQN, Krusell–Smith with deep learning, and the sequence-space reformulation.